

## 2026-04-21 : Briefing Document

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## Exploratory Data Analysis (EDA) and Data Quality Briefing

### Executive Summary

This briefing document synthesizes the core principles and findings of an exploratory data analysis (EDA) conducted using a clinical dataset of blood parameters and patient ailments, as an example. The analysis serves as a foundational step in the machine learning lifecycle.

### Critical Takeaways:

- **Data Quality Crisis:** The initial dataset of 2,370 observations contained 2,282 duplicate rows, leaving approximately 100 unique observations for modeling.
- **Methodological Rigor:** EDA must be guided by domain expertise rather than blind mathematical application. Statistical rules (like the **1.5 x IQR** outlier rule) are guidelines, not absolute laws.
- **Metric Selection:** In cases of data imbalance—such as the underrepresentation of specific ailments like thrombocytosis—accuracy is a misleading metric. The F1 score is the preferred metric for assessing model performance.
- **Feature Distribution:** The clinical parameters in this dataset do not follow standard normal distributions; instead, they appear random and exhibit low pairwise correlation, suggesting a complex, non-linear relationship between features.

## 1. The Machine Learning Framework: $Y = f(X)$

The objective of the analysis is to identify the function ( $f$ ) that explains the relationship between independent variables ( $X$ ) and the dependent variable ( $Y$ ).

- **Dependent Variable ( $Y$ ):** The ailment (e.g., anemia, diabetes, heart disease). This is a **nominal** discrete variable used for classification.
- **Independent Variables ( $X$ ):** 24 blood parameters (labeled P1 to P24). These are **continuous** variables (interval or ratio scales, mostly ratio-scales since the value ZERO is a meaningful value in most of these parameters).
- **The Goal:** Predict ailments based on blood test results, creating a diagnostic capability through machine learning.

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## 2. Data Integrity and Cleaning Findings

A rigorous examination of the "patient\_data.csv" file revealed several critical integrity issues that must be addressed before model training.

### 2.1 Duplicate Observations

The analysis identified a "shocker": 2,282 out of 2,370 rows were exact duplicates.

- **Action taken:** These rows were dropped to prevent biased weightage.
- **Inference:** From a domain perspective, it is virtually impossible for two different blood reports to be identical down to the last decimal across 24 parameters. This suggests a data entry or collection error.

### 2.2 Missing Values

The dataset contained sporadic missing values across several columns:

- **Affected Columns:** P3, P4, P5, P6, P15, P16, and the Ailment column.
- **Strategic Approach:** If a single row contains a significant number of missing values, the entire row should be dropped, as it provides insufficient information for analysis.

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## 3. Statistical Analysis and Visualization

The analysis utilized various statistical tools to understand the nature of the data.

### 3.1 Levels of Measurement

| Variable Category | Level          | Nature     | Analysis Tools                       |
|-------------------|----------------|------------|--------------------------------------|
| Ailments          | Nominal        | Discrete   | Bar Plots, Frequency Charts          |
| Blood Parameters  | Interval/Ratio | Continuous | Histograms, Box Plots, Scatter Plots |

### 3.2 Outlier Detection (The 1.5 x IQR Rule)

Outliers are identified using the Interquartile Range (IQR), defined as  $Q3 - Q1$ .

- **Lower Limit:**  $Q1 - (1.5 \times IQR)$
- **Higher Limit:**  $Q3 + (1.5 \times IQR)$
- **Findings:** The current dataset yielded zero outliers. However, the analysis warns against the "blind application" of these limits. Real-world occurrences (e.g., a high-income individual in a low-income village) may be statistically outliers but factually representative of reality.

### 3.3 Data Distribution

Visual analysis through histograms and Kernel Density Estimates (KDE) revealed:

- **Lack of Normality:** Most parameters did not follow a normal or Gaussian distribution.

- **Randomness:** Scatter plots indicated a high degree of randomness in the sequence of data, with no immediately visible patterns.
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#### 4. Class Imbalance and Metric Selection

A significant challenge identified during EDA is **Data Imbalance** within the ailment categories.

- **Observations:** While most ailments had 15–20 unique observations, "thrombocytosis" had only three.
  - **Impact:** Machine learning algorithms learn from examples. Underrepresented classes like thrombocytosis will likely be misclassified (often as heart disease) because the model lacks sufficient data to learn their specific characteristics.
  - **Metric Shift:** Accuracy provides a "false picture" in imbalanced datasets. A model could incorrectly classify all instances of a rare disease but still maintain 98% accuracy. **F1 Score** (the harmonic mean of precision and recall) must be the primary metric for decision-making.
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#### 5. Feature Relationships and Dimensionality

The analysis explored the "Curse of Dimensionality"—the principle that increasing the number of columns can make solving problems more difficult.

##### 5.1 Correlation Heat Map

A heat map was used to establish the relationship between blood parameters.

- **Findings:** The dataset exhibited extremely low correlation. Most values ranged between 0.2 and -0.3.
  - **Implication:** Since variables are not highly correlated, they likely provide distinct information. In general, however, if two variables were highly correlated, one could be dropped to simplify the model without losing information.
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#### 6. Analytical Mindset and Best Practices

The briefing concludes with several guiding principles for data science professionals:

"No decision is purely mathematical in nature in data science... these are only guidelines. Your final decision should be based on the domain."

- **Avoid Blind Automation:** While LLMs and Python scripts can generate diagnostics, they can "disobey" sequence instructions or hallucinate. Human oversight is mandatory.
- **Domain Justification:** Before dropping an outlier or a duplicate, an analyst must convince themselves that the action is correct from a domain perspective.
- **Contextual Relevance:** Data dropped in one context (e.g., routine analysis) might be the most critical data in another (e.g., identifying rare medical anomalies). Analysts must "keep in mind" what has been excluded.

# Exploratory Data Analysis and Machine Learning Fundamentals: Study Guide

This study guide provides a comprehensive review of the principles of Exploratory Data Analysis (EDA) and the initial steps of the machine learning lifecycle, as discussed in the provided lecture transcript. It covers data classification, outlier detection, visualization techniques, and the challenges of data imbalance.

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## Part 1: Short-Answer Quiz

**Instructions:** Answer the following questions in 2–3 sentences based on the information provided in the source context.

1. What is the fundamental definition of machine learning as described in the text?
  2. How are "Nominal" and "Ordinal" variables distinguished from "Interval" and "Ratio" variables in terms of data types?
  3. Why is business understanding considered the starting point of the data science lifecycle?
  4. What criteria should a data scientist use before deciding to drop duplicate rows in a dataset?
  5. Explain the relationship between a box plot and a histogram regarding data distribution.
  6. How is the Interquartile Range (IQR) calculated, and what does it represent?
  7. What are the mathematical boundaries used to identify a "nominal" outlier in a box plot?
  8. What is "Data Imbalance," and how does it affect a machine learning model's performance?
  9. Why is the F1 score often a more reliable metric than accuracy in classification problems?
  10. What is the purpose of a Correlation Heat Map in exploratory data analysis?
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## Part 2: Quiz Answer Key

1. **Fundamental Definition:** In many supervised learning problems, machine learning can be framed as learning a function that maps inputs  $X$  to outputs  $Y$  ( $Y \approx f(X)$ ). More broadly, machine learning refers to algorithms that learn patterns from data to make predictions, decisions, or discover structure.
2. **Distinguishing Data Types:** Nominal and ordinal variables are classified as discrete data, used primarily to distinguish or order categories without numerical value. Interval and ratio variables are classified as continuous data, representing real numbers and decimal values that allow for a broader range of statistical calculations.
3. **Business Understanding:** Everything in the data science lifecycle begins with business understanding to establish clear objectives and goals. This process ensures the data scientist knows exactly what capability they are trying to create, such as predicting specific ailments based on blood test parameters.
4. **Dropping Duplicates:** Before taking action, a researcher must convince themselves through domain knowledge that the duplicates are errors rather than reflections of reality. In some

fields, like blood testing, identical reports down to the last decimal are considered a physical impossibility, justifying their removal.

5. **Box Plots vs. Histograms:** Both tools visualize data distribution, but while a histogram shows frequency, a box plot displays the density of observations within quartiles. A skewed box plot—where the median is not centered—indicates a skewed distribution in the corresponding histogram.
6. **IQR Calculation:** The Interquartile Range is calculated as  $Q3 - Q1$ , representing the difference between the 75th and 25th percentiles. This range contains the middle 50% of the observations in a sorted dataset.
7. **Outlier Boundaries:** Boundaries are established using the 1.5x IQR rule: the Lower Limit is  $Q1 - (1.5 \times IQR)$  and the Higher Limit is  $Q3 + (1.5 \times IQR)$ . Observations falling outside these "whiskers" are flagged as **potential outliers** for further investigation.
8. **Data Imbalance:** Data imbalance occurs when one class (e.g., a specific disease) is significantly underrepresented compared to others. This prevents machine learning algorithms from learning that class effectively, often leading the model to incorrectly classify rare cases as more common ones.
9. **F1 Score vs. Accuracy:** Accuracy can be misleading in imbalanced datasets because a model could ignore a rare class entirely and still achieve a high percentage of correct predictions. The F1 score, which combines precision and recall, provides a more realistic picture of a model's performance across all classes.
10. **Correlation Heat Map:** A Heat Map is a visual tool that uses color-coding to show the correlation coefficients between every pair of columns in a dataset. It helps identify redundant variables, allowing researchers to reduce the number of columns and avoid the "curse of dimensionality."

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### Part 3: Essay Questions

**Instructions:** Use the concepts from the source to develop detailed responses to the following prompts. (Answers not provided).

1. **The Subjectivity of Data Cleaning:** Discuss the statement "No decision is purely mathematical in nature in data science." Use the examples of outliers and duplicate data to explain why domain expertise is required to supplement mathematical guidelines.
2. **The Anatomy of a Box Plot:** Describe how to construct a box plot from a raw, unsorted list of continuous data. Explain the significance of the median ( $Q2$ ), the quartiles ( $Q1$ ,  $Q3$ ), and how the "whiskers" relate to the standard definition of an outlier.
3. **Visualization Strategies in EDA:** Compare and contrast Scatter Plots, Pair Plots, and Histograms. Under what circumstances would one be more useful than the others for identifying patterns or randomness in a dataset?
4. **Managing Underrepresented Classes:** Analyze the consequences of data imbalance in a medical diagnostic context. Evaluate the potential solutions of "downsampling" versus "upsampling" and explain why one might be preferred over the other based on the size of the dataset.

5. **The Curse of Dimensionality:** Using the Correlation Heat Map as a reference point, explain why increasing the number of columns (X variables) does not always lead to a better machine learning model. How can a researcher use correlation data to simplify their model?
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#### Part 4: Glossary of Key Terms

| Term                             | Definition                                                                                                                                                 |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Accuracy</b>                  | A performance metric representing the percentage of total correct predictions; can be misleading in imbalanced datasets.                                   |
| <b>Box Plot</b>                  | A standardized way of displaying the distribution of data based on a five-number summary: minimum, first quartile, median, third quartile, and maximum.    |
| <b>Continuous Variable</b>       | A variable that can take any value within a range (real numbers/decimals), typically associated with Interval or Ratio levels of measurement.              |
| <b>Correlation Coefficient</b>   | A numerical measure that indicates the strength and direction of a relationship between two variables, ranging from -1 to +1.                              |
| <b>CRISP-DM</b>                  | A standard model (Cross-Industry Standard Process for Data Mining) used as a guideline for data science projects.                                          |
| <b>Data Imbalance</b>            | A situation in a classification problem where the number of observations for different classes is not roughly equal.                                       |
| <b>Discrete Variable</b>         | A variable with a finite number of distinct values, typically associated with Nominal or Ordinal levels of measurement.                                    |
| <b>F1 Score</b>                  | The harmonic mean of precision and recall; used as a preferred metric in classification problems involving imbalanced data.                                |
| <b>Heat Map</b>                  | A graphical representation of data where individual values contained in a matrix are represented as colors.                                                |
| <b>Interquartile Range (IQR)</b> | The distance between the first and third quartiles ( $Q3 - Q1$ ), representing the spread of the middle 50% of data.                                       |
| <b>KDE Plot</b>                  | Kernel Density Estimate; a way to estimate the probability density function of a continuous random variable, resulting in a smooth curve over a histogram. |
| <b>Median (Q2)</b>               | The middle value in a sorted dataset; the 50th percentile.                                                                                                 |
| <b>Nominal Variable</b>          | A discrete variable used to distinguish between categories that have no inherent ordering (e.g., names of ailments).                                       |
| <b>Outlier</b>                   | An observation that lies an abnormal distance from other values in a random sample from a population.                                                      |
| <b>Pair Plot</b>                 | A grid of scatter plots showing the relationship between every possible pair of variables in a dataset.                                                    |

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| <b>Quartile</b> | Values that divide a sorted dataset into four equal parts; Q1 is the 25th percentile and Q3 is the 75th percentile. |
| <b>Skewness</b> | A measure of the asymmetry of the probability distribution of a real-valued random variable.                        |

| Question                                                                                      | Answer                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What are Y and X in this particular dataset?                                                  | <b>Y represents the ailment</b> (dependent variable), and <b>X represents the blood parameters</b> or P-values (independent variables).                                                                                                                                               |
| Would the "ailment" column be considered nominal or ordinal?                                  | <b>It is a nominal variable</b> because there is no logical ordering between the different diseases.                                                                                                                                                                                  |
| Should duplicate rows in the dataset be kept or dropped?                                      | <b>They should be dropped.</b> Keeping them inappropriately increases the weightage of those specific observations, and it is practically impossible for real blood reports to have exactly identical decimal values.                                                                 |
| Is it good or bad that the LLM performed an outlier analysis earlier than instructed?         | <b>It is both good and bad.</b> It is common practice to perform outlier analysis early, but the LLM hallucinated by disobeying explicit prompt instructions, highlighting the need to verify automated tools.                                                                        |
| What can be concluded from the scatter plot of the 24 variables?                              | <b>The variables appear completely random</b> and do not seem to follow any discernible pattern.                                                                                                                                                                                      |
| What is the conclusion regarding the distribution of the columns?                             | <b>The data is not normally distributed</b> and does not appear to possess any specific underlying distribution or pattern.                                                                                                                                                           |
| Will the data yield a good model as is?                                                       | <b>Likely not.</b> The data will probably need to be processed or transformed before a good model can be created.                                                                                                                                                                     |
| Will the model be a regression or classification model?                                       | <b>It will be a classification model</b> because the dependent variable (the ailment) is a discrete category.                                                                                                                                                                         |
| What is the median?                                                                           | <b>It is the middle value</b> , or 50th percentile, that evenly divides the sorted dataset in half.                                                                                                                                                                                   |
| Why might Q2 (the median) not sit exactly in the middle of a box plot's interquartile range?  | <b>It indicates an asymmetric, skewed distribution</b> where the density of the data points is much higher in one half of the box than the other.                                                                                                                                     |
| Doesn't the peak of a normal distribution represent the mean, while Q2 represents the median? | <b>Yes, the peak represents the mean.</b> Typically, in a real life dataset, the mean and median are not identical values. Actual data distributions are rarely "pure"<br><br>In the case of pure (theoretical) <b>Normal Distribution</b> the mean, median, and the mode – coincide. |



|                                                                                                   |                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Should we look at the histogram before outlier analysis to understand skewness?                   | It is not strictly necessary, as a box plot inherently includes the distribution information that a histogram provides.                                                                                                                                  |
| How does a box plot handle identifying outliers if a distribution has a heavy tail on both sides? | The box plot establishes rigid boundaries (whiskers) based on the Q1 and Q3 limits. Any value falling outside of these limits is flagged as an outlier, regardless of how long the tail extends.                                                         |
| How do we identify outliers for a categorical column (like country or gender)?                    | Box plots and outlier identification are meant for continuous variables, not nominal ones. To analyze by category, you apply a filter for that specific category (e.g., a specific country) and create box plots for the continuous variables within it. |
| If there is a correlation in a pair plot, will it show a clear pattern?                           | Generally yes, but the lack of an obvious visual pattern does not strictly guarantee that there is zero correlation.                                                                                                                                     |
| Is there a strict percentage or thumb rule for when a class is considered "imbalanced"?           | No, there is no strict numeric rule. It heavily depends on the context of the domain and proper justification.                                                                                                                                           |
| Why is the F1 score a better metric than accuracy for imbalanced data?                            | If a minority class is misclassified, the F1 score will accurately drop to reflect that failure, whereas accuracy might incorrectly remain as high as 97-98%.                                                                                            |
| If row A and row B are duplicates, can we just drop one and keep the other?                       | Yes, that is precisely what dropping duplicates entails—retaining one unique copy of the observation while dropping the identical redundancies.                                                                                                          |
| If only a single feature is different between two rows, is it still a duplicate?                  | No, if even one feature differs, the row is considered a distinct observation.                                                                                                                                                                           |
| If two parameters have identical distribution shapes, does that mean they are correlated?         | No. A histogram only details the distribution of values within a single, isolated column; it does not measure the relationship between two separate columns.                                                                                             |
| Why does a pair plot show a pattern when plotting a column against itself?                        | The diagonal in a pair plot is not a scatter plot. It is actually a frequency plot (distribution/histogram) uniquely characterizing that single column.                                                                                                  |
| Why wouldn't a variable plotted against itself follow a normal distribution?                      | There is no rule stating a variable must be normal. The data could naturally follow any other distribution, such as a uniform distribution.                                                                                                              |

|                                                          |                                                                                                                                                                                   |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How does a box plot represent a wavy (multimodal) graph? | A box plot is completely independent of the visual shape of the distribution. It strictly marks the median and shows the density or spread of the middle 50% of the observations. |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Questions That Were Not Answered

- **"What is the reason for this 1.5 times [for IQR outlier bounds]? Why is it not 1.25? Or why is it not 1.75?"** The faculty explicitly left this unanswered, assigning it to the students as a problem to think about.
- **"Is the concept of outlier applicable to, let us say, a nominal or an ordinal variable in this particular sense?"** The faculty posed this to the students with the instruction to "Think about it," though they hinted that nominal variables lack a mean or median.
- **"Can we use the remaining counts as a test data [in a downsampling scenario]?"** The faculty declined to answer this directly, stating that the test-train split is a critical topic that would be specifically covered in the next class.
- **"Do we know what are the row index?"** A student asked this in the chat, but the instructor never addressed it.
- **"What do you mean by asymmetric distribution?"** The faculty asked a student this question, but the student never answered; the faculty then proceeded to explain it themselves.

Would you like me to create flashcards or a study guide based on the concepts discussed in this Q&A table?

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